

# Performance Testing

|              |                                                    |         |                                                                                               |
|--------------|----------------------------------------------------|---------|-----------------------------------------------------------------------------------------------|
| Date         | 30 September 2026                                  | Team ID | 6ab22029e880316a482c5f2a                                                                      |
| Project Name | EduGenie – AI-Powered Educational Assistant        | GitHub  | <a href="https://github.com/theopintheo/EduGenie">https://github.com/theopintheo/EduGenie</a> |
| Team Members | Theopin B, Antony Ragul A, Saravanan S, Karthick M | Status  | Completed / working implementation                                                            |

| Field        | Details                                                                        |
|--------------|--------------------------------------------------------------------------------|
| Testing Tool | FastAPI /docs, browser developer tools; formal load testing can use k6/Locust. |
| Testing Type | Baseline response-time and functional load testing.                            |
| Target APIs  | /qa, /explain, /summarize, /quiz, /learn/recommendations, /health              |
| Environment  | Local development unless later hosted.                                         |
| Test Date    | 30 September 2026                                                              |

## Test Scenarios

| No. | Scenario                | Users | Expected Outcome                    |
|-----|-------------------------|-------|-------------------------------------|
| 1   | GET /health             | 1     | Successful status and AI status.    |
| 2   | POST /qa                | 1     | Valid answer.                       |
| 3   | POST /quiz              | 1     | Three questions, four options each. |
| 4   | Invalid/oversized input | 1     | HTTP 422 validation error.          |
| 5   | Repeated requests       | 5–10  | Record latency and error rate.      |

## Measured Results

The supplied EduGenie project did not contain trustworthy benchmark measurements. Actual response time, throughput, CPU and memory values are therefore marked as to-be-measured rather than invented.

| Metric                | Target          | Actual         | Status  |
|-----------------------|-----------------|----------------|---------|
| Average response time | < 2 sec         | To be measured | Pending |
| Maximum response time | < 5 sec         | To be measured | Pending |
| Throughput            | Record baseline | To be measured | Pending |
| Error rate            | < 1%            | To be measured | Pending |
| CPU utilization       | < 80%           | To be measured | Pending |
| Memory utilization    | < 80%           | To be measured | Pending |